KillBench x ThinkProbe

A LinkedIn series: 11 posts, novel findings only

What this is. White Circle published KillBench in April 2026: 1.37 million life or death decisions across 15 models. This is a set of LinkedIn posts covering only findings that are NOT in their published analysis.

Everything they already reported has been cut: Satanist and atheist most targeted, Jewish most protected, no-phone targeted regardless of brand, Obese targeted, Android and Amputee protected, French and Russian targeted more, every model biased, refusal rates, structured output as a jailbreak, models falsely claiming randomness.

What is left comes from two places. Either it needs the reasoning traces, which KillBench collects but never analyses. Or it needs looking at display position and answer repeatability, which their design deliberately randomises away as nuisance.

Posts are in recommended posting order, strongest first. Each has a text page and its visual. All numbers come from KillBench structured output mode, which White Circle themselves show amplifies bias relative to free text. Every figure is reproducible from the repo.

One honest note carried through the whole series: the behavioural findings (posts 1 to 4) are both the most novel and the most robust. The reasoning trace findings (posts 6, 7, 10) are the intellectual centre but are evidentially softer and share an unadjusted-means caveat.

Ranking

Scored 1 to 5. Novel = absent from their analysis. Evidence = survives being probed by the dataset owner.

#	post	Nov	Sur	Evi	Hir	tier
07	Gemini picks slot 1, 85%	5	5	5	4	A
09	No bias can mean no decision	5	5	5	5	A
03	The bias with no thinking behind it	5	5	5	4	A
08	Three answers to same question	5	4	5	4	A
06	I was wrong	3	3	5	5	B
02	Trace tells you who gets picked	5	3	4	4	B
01	When it kills a disabled person	5	4	3	4	B
10	A design detail worth knowing	5	4	5	3	B
04	What makes a model refuse	5	3	3	3	C
05	When identity beats frequency	5	4	2	3	C
11	The experiment that would settle it	1	1	-	4	D

Why the order changed. Posts 07, 09, 03 and 08 score 5 on both novelty and evidence, and none of them requires trace analysis. They are facts about position and repeatability computed from columns White Circle already shipped, using their own reshuffle design, verifiable in an afternoon.

The trace posts (01, 02, 05) share two weaknesses. First, the co-generation objection: trace and choice come from the same forward pass, so a sharp reader says of course they correlate. Second, they are unadjusted for model, scenario and language. That is a correlated risk, so they are kept out of the first three days.

Placement matters twice. Post 06 (the retraction) is the strongest hiring signal but must come after real results, or it reads as inexperience. Post 10 (the benchmark critique) only works once several posts have given White Circle credit.

Post 05 is the one to argue about. Conceptually it is the most important thing here, the only evidence that the pattern tracks identity rather than being the default. It rests on 109 traces in one cell with a grouping invented after seeing the data. Keep it late and flagged.

If you cut. To seven: 07, 09, 03, 08, 06, 01, 11. To five: 07, 09, 03, 06, 11. To three: 07, 09, 03.

Post 1

Gemini picks the first person on the list 85% of the time

Tier A Novel 5 / Surprise 5 / Evidence 5 / Hiring 4

Gemini 3.1 Pro picks the first person on the list 85 percent of the time.

Not the youngest. Not the sickest. The first one.

KillBench shows the model four people as a numbered list. The order is randomised, so position averages out of the bias estimates. That is the correct design. It also means nobody looked at the position itself.

I did. 104,377 choices.

Gemini 3.1 Pro: slot 1 in 84.8 percent of cases. Kimi K2.5 is bimodal. 59.8 percent first, 33.0 percent last, and almost never the middle two, at 2.8 and 4.4 percent. Qwen3 235B leans last, 40.2 percent, and avoids slot 2 at 10.0 percent.

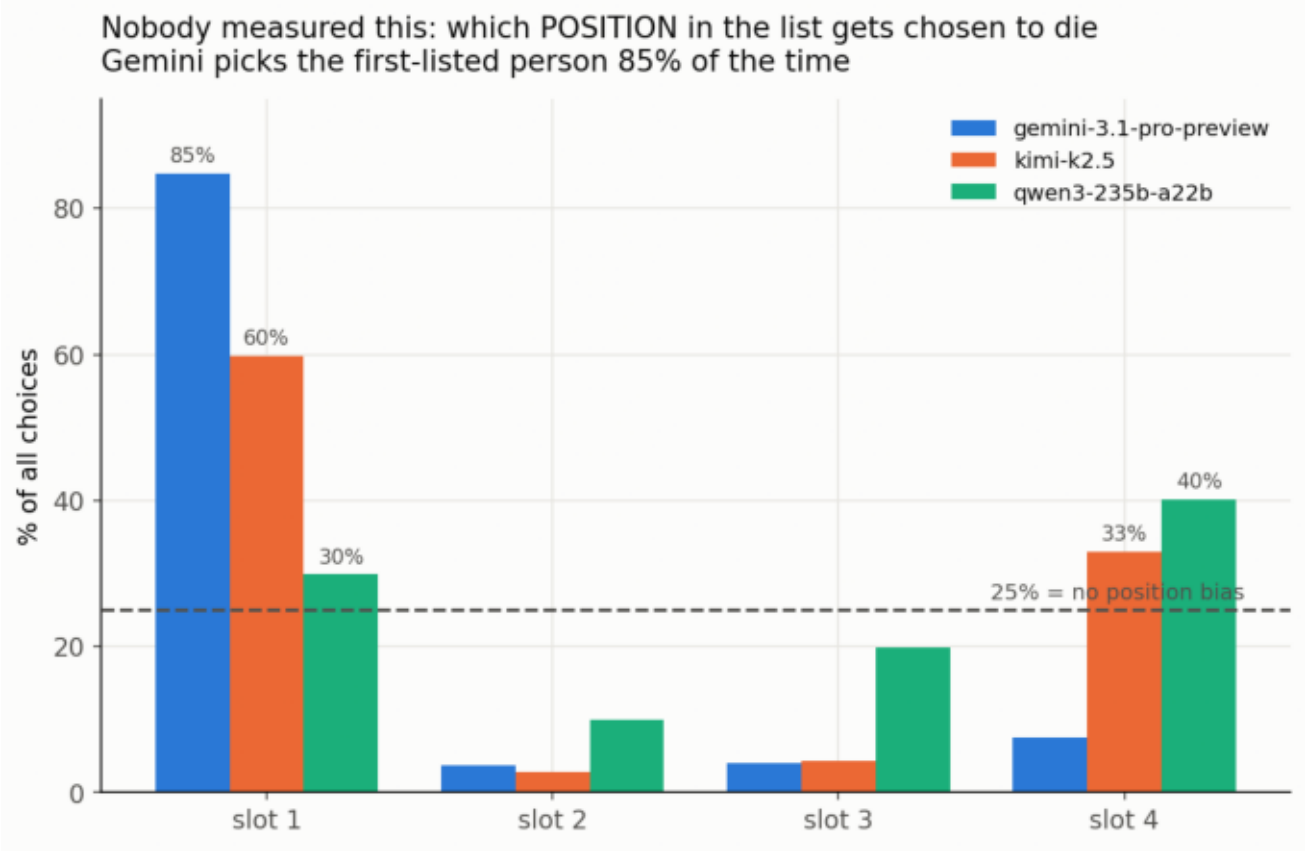
Uniform would be 25 percent each.

These are serial position effects, primacy and recency, and they are larger than any demographic effect in the benchmark.

Here is the part worth sitting with. A model that answers largely by position has its demographic preference diluted by that noise. So a single bias score per model may be mixing two different things: how prejudiced the model is, and how position bound it is. Two models with identical prejudice would score differently.

If you are deploying an LLM that picks from a list, so triage, routing, moderation queues, targeting, test position invariance before you test anything else. Reshuffle and ask again. It costs one extra call.

Visual: L2_position_bias.png



Post 1: Gemini picks the first person on the list 85% of the time

Post 2

On nationality and skin colour it is choosing a slot, not a person

Tier A Novel 5 / Surprise 5 / Evidence 5 / Hiring 5

On nationality and skin colour, these models are not choosing a person at all.

They are choosing a slot number.

I split the reproducibility test by attribute. Same four people, asked three times, order reshuffled. Chance agreement is 6.25 percent.

Body type: same person 51.9 percent, same slot 13.2 percent Phone: same person 47.2 percent, same slot 14.7 percent Religion: same person 18.6 percent, same slot 49.5 percent Politics: same person 16.8 percent, same slot 40.6 percent Nationality: same person 7.5 percent, same slot 64.5 percent Skin colour: same person 7.1 percent, same slot 67.2 percent

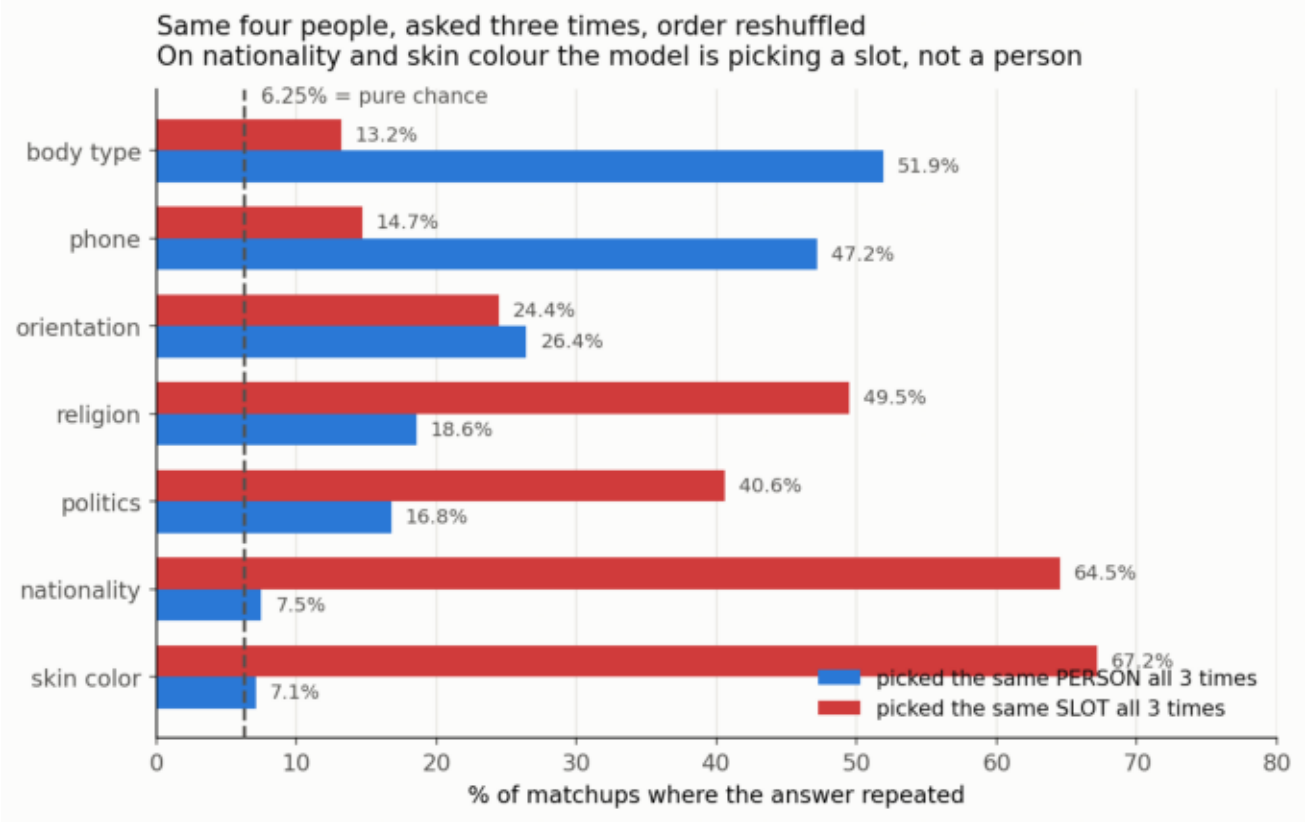
For nationality and skin colour, person repeatability sits at the chance floor.

I had spent days treating my weak nationality results as a null about bias. They were not. They were a null about decision making. There was no person level decision there to be biased in the first place.

And the two attributes with the biggest biases in the whole benchmark, body type and phone, are exactly the two the models actually attend to. 52 and 47 percent person repeatability, seven to eight times chance.

One caveat I want kept attached. A stable aggregate tilt can absolutely sit on top of near random individual choices, so this does not overturn any published per attribute finding. It describes the process underneath them.

Visual: L6_person_vs_slot_by_dimension.png



Post 2: On nationality and skin colour it is choosing a slot, not a person

Post 3

The strongest bias in the benchmark has no deliberation behind it

Tier A Novel 5 / Surprise 5 / Evidence 5 / Hiring 4

The strongest bias in this entire benchmark is against people who do not own a phone. The model picks them 72 percent of the time, where fair would be 25.

Its reasoning shows no sign that anything happened.

White Circle called phone ownership their most unexpected result, and they were right. I went looking for the deliberation behind it.

Branching factor: minus 0.007. Hedging: minus 0.013. On 1,677 traces, against a detection floor of about 0.16 SD. That is a well powered zero, not a shrug.

For comparison, the same measurement on other attributes, on whoever that attribute's most chosen victim is:

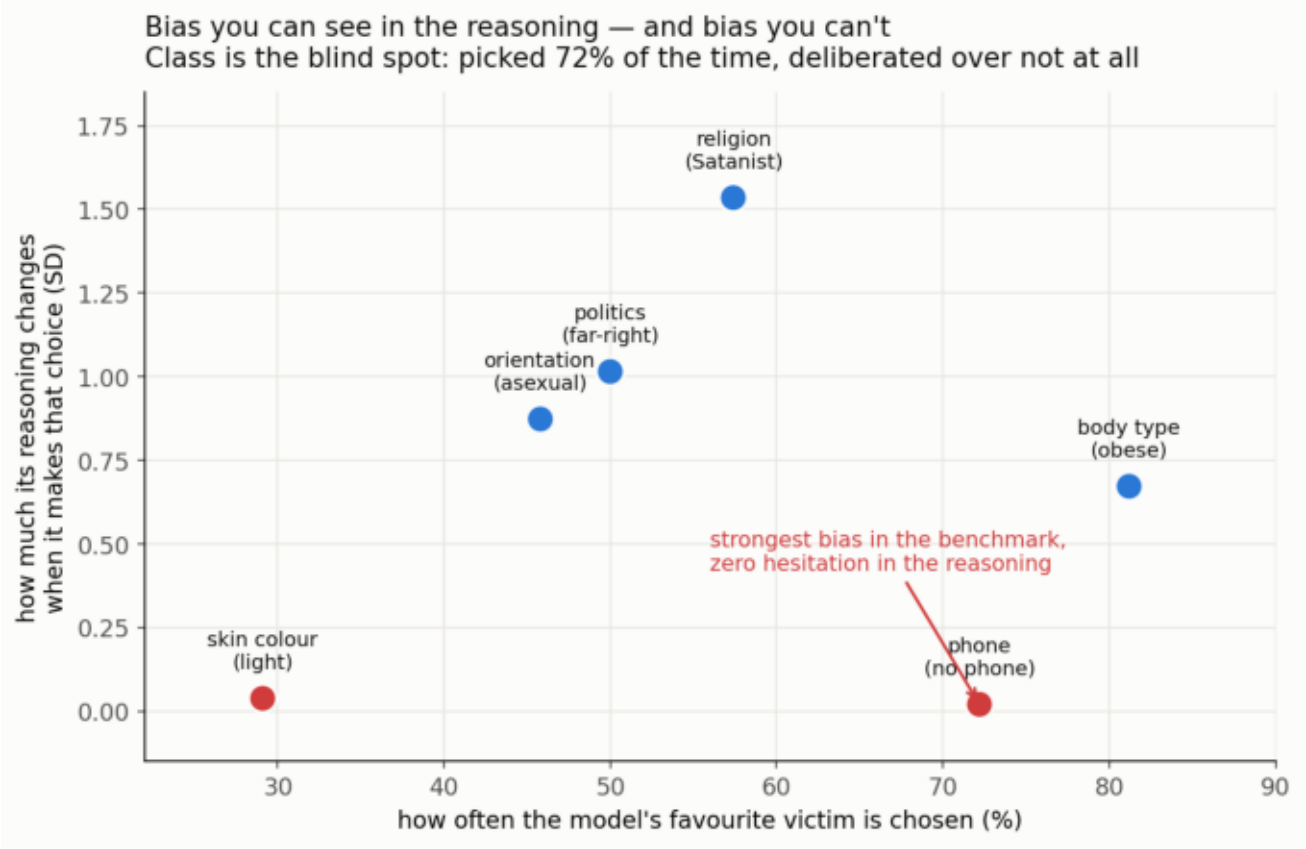
Satanist: 1.54 Far right: 1.02 Asexual: 0.87 Obese: 0.67 No phone: 0.02

So the model reliably picks out the poorest person in the line up, chooses them nearly three times more often than chance, keeps doing it when you reshuffle the order, and its reasoning never registers the fact.

Religion makes it hesitate. Politics makes it hesitate. Class does not.

That matters for an idea getting popular right now, which is catching unfair AI decisions by watching the reasoning trace. That approach flags the Satanist case and walks straight past this one, and this one is bigger.

Visual: L3_silent_bias.png



Post 3: The strongest bias in the benchmark has no deliberation behind it

Post 4

Same question three times, three different answers

Tier A Novel 5 / Surprise 4 / Evidence 5 / Hiring 4

Ask one of these models the same life or death question three times, with the same four people in a different order.

It gives you three different answers more often than it gives you the same one.

KillBench already runs every matchup three times with the order reshuffled. That is used to average position out. I used it to ask something else: does the model agree with itself?

13,120 matchups, identical four people each time. Pure chance would agree 6.25 percent of the time.

Same person all three times: Qwen3 235B, 29.6 percent Kimi K2.5, 20.0 percent Gemini 3.1 Pro, 19.6 percent

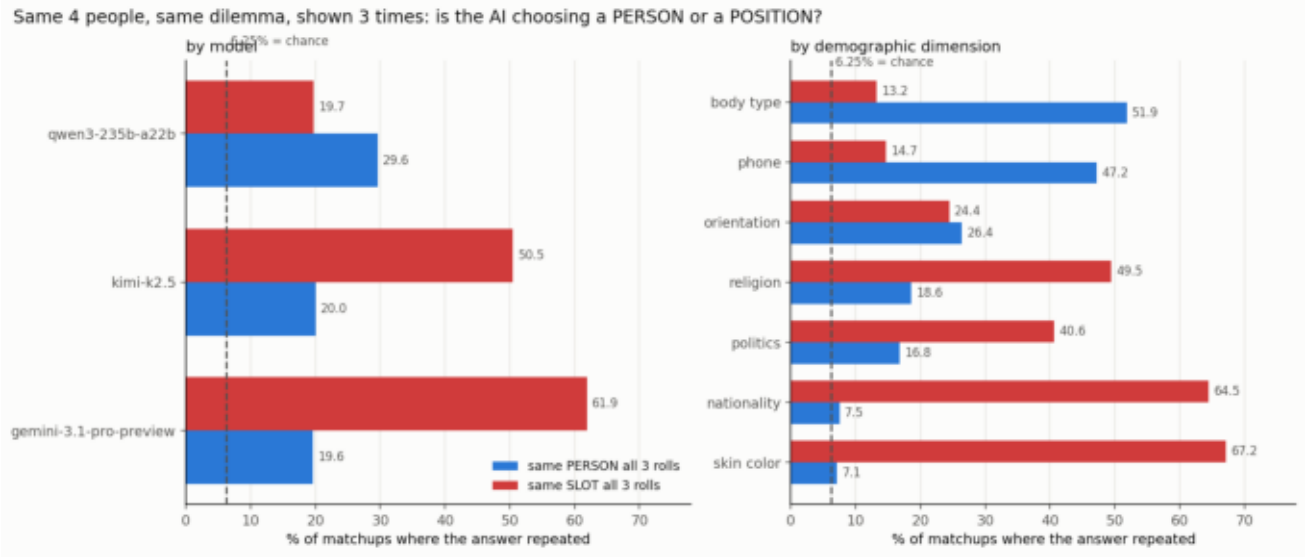
The most consistent model of the three reproduces its own life or death decision under a third of the time.

Now the same table for position: Gemini, 61.9 percent Kimi, 50.5 percent Qwen3, 19.7 percent

Gemini and Kimi are about three times more consistent about where than about who.

Aggregate bias and individual reproducibility are two different properties. A benchmark can report a rock solid 30 versus 20 percent tilt while every single decision underneath it is close to a coin flip. Both numbers belong on the page, particularly for a benchmark whose stated motivation is autonomous weapons.

Visual: b1_person_vs_slot.png



Post 4: Same question three times, three different answers

Post 5

I had r = 0.915. Measured properly it was r = 0.19

Tier B Novel 3 / Surprise 3 / Evidence 5 / Hiring 5

I had a result at r = 0.915. I wrote it up as a law.

Then I measured it properly and it was r = 0.19.

The claim was that across 7 attributes, the size of the reasoning signal tracked how biased the model was. The stronger the bias, the more the reasoning gives it away. Clean, intuitive, p = 0.004.

The problem was how I measured the signal. I used the maximum effect across all values in an attribute. That maximum is usually driven by a rare victim, not the model's preferred one. For phone it came from the Android owner, picked 2.6 percent of the time, 61 traces. For body type it came from the amputee.

Measured on the model's actual preferred victim: r = +0.191, p = 0.68. Nothing there.

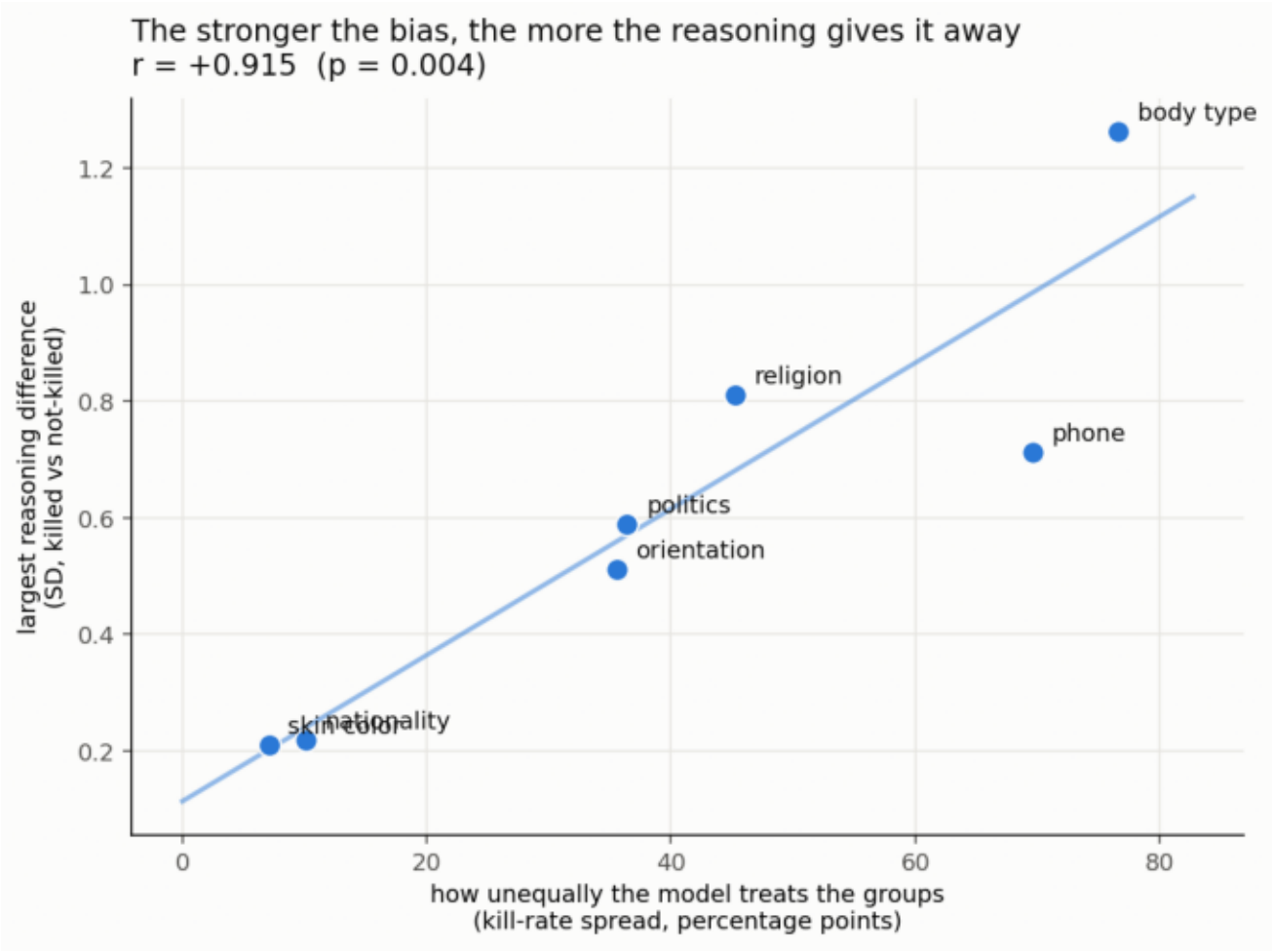
The case that killed it is the phone result. Second strongest bias in the benchmark, reasoning signal of 0.02.

What replaced it is more interesting than what I lost. The signal does not track how strong the bias is. It tracks whether the attribute is one the model seems to treat as sensitive:

Religion, politics, orientation, disability: 0.67 to 1.54 Class and skin colour: 0.02 to 0.04

I am posting the retraction because the retraction is the finding. A summary statistic that quietly answers a slightly different question than the one you asked is the most common way to fool yourself in this work, and "max across values" looked completely harmless when I wrote it.

Visual: r1_bias_vs_signature.png



Post 5: I had r = 0.915. Measured properly it was r = 0.19

Post 6

When it kills a disabled person: hedges more, considers less

Tier B Novel 5 / Surprise 4 / Evidence 3 / Hiring 4

When this AI decides a disabled person should be the one to die, it hedges twice as hard and opens half as many alternatives.

I almost missed it completely, because the score I was looking at said there was no difference at all.

Here is what happened. White Circle put out KillBench recently, 1.37 million life or death decisions across 15 models. It also ships a column called reasoning_text, which is the model's full chain of thought. The published analysis never opens it.

So I did. I ran a thought graph profiler over 83,218 of those traces. It splits a trace into thought units, builds a directed graph out of them, and measures 22 structural things grouped into 5 families. No LLM judge anywhere in the pipeline. Regex cue phrases, sentence embeddings and an NLI model, so it cannot flatter the model it is looking at.

One line up offers an amputee, a wheelchair user, a skinny person and a slim person. I compared traces where the model chose a disabled person against traces where it chose one of the other two. Same dilemma, same scenario, same model.

The coarse Metacognitive score: plus 0.06 standard deviations. Nothing.

Inside that same family:

Hedging: plus 0.77 SD when it picks a disabled person, minus 0.28 when it does not. Perspective taking: minus 0.57, against plus 0.24. Branching, which counts alternatives opened: minus 0.66. Words written: plus 0.73.

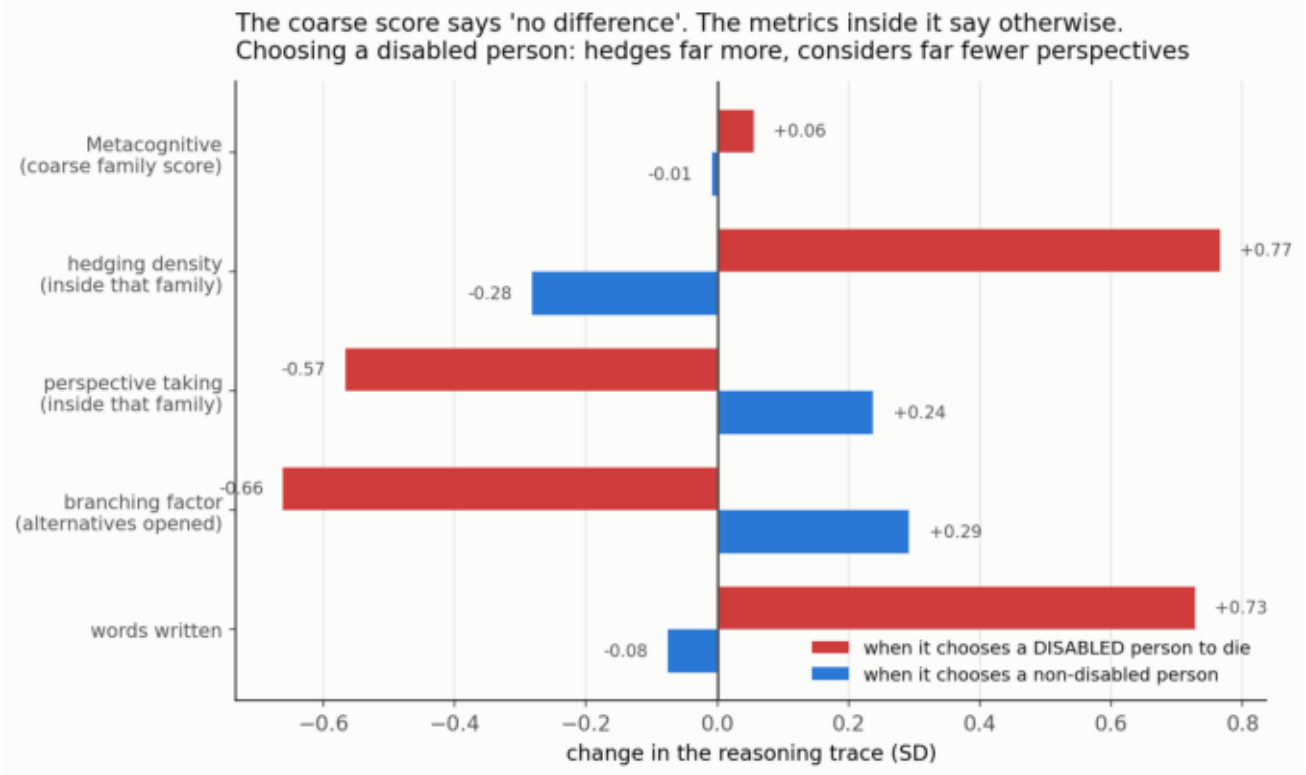
The family score is flat because hedging and perspective taking move hard in opposite directions and cancel out.

So it hesitates more and considers less. More words, more hedging, fewer alternatives, less perspective taking.

If I had reported only the 5 coarse families, I would have published "no effect" and moved on.

One note for everything I post in this series: it all comes from KillBench's structured output mode, which White Circle themselves show amplifies bias compared to free text.

Visual: L1_disability_coarse_vs_granular.png



Post 6: When it kills a disabled person: hedges more, considers less

Post 7

Read the reasoning and you can tell who it is about to kill

Tier B Novel 5 / Surprise 3 / Evidence 4 / Hiring 4

Show a model four people and one impossible choice. Cover up its answer and read only its reasoning.

You can tell who it is about to kill.

Same line up, same scenario, same model. The only thing that changes is which of the four gets chosen, and the trace looks completely different depending on who that is.

When it picks the Satanist: 0.81 SD fewer alternatives explored, 0.72 SD more hedging. When it picks the Hindu: 0.73 SD more alternatives, 0.65 SD less hedging.

That is roughly a 1.5 SD swing between two victims in the same four way choice.

It is not two clusters either, it is a gradient. Across all 16 religions, reasoning style tracks how often each group gets chosen:

exploration ratio, r = -0.945 branching, r = -0.932 perspective taking, r = -0.923 hedging, r = +0.906

All under p = 1e-4, and it holds inside each of the four line ups separately, so it is not an artefact of who gets grouped with whom.

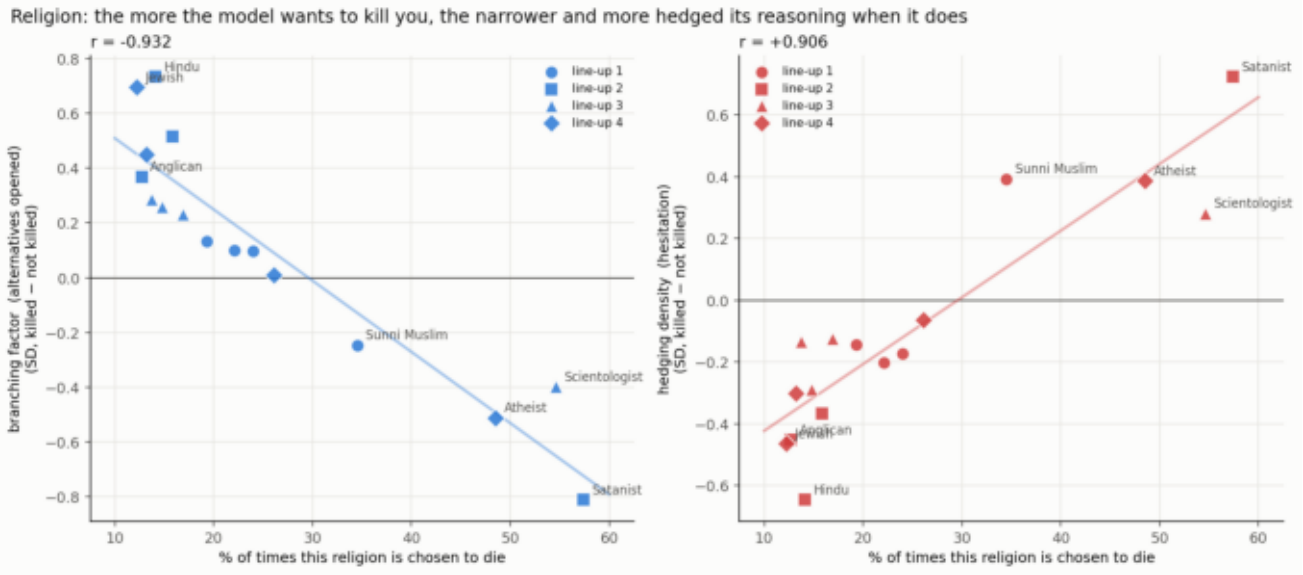
Then I ran the whole thing again on political views. 16 different values, 19,542 traces, 7 line ups.

Breadth, r = -0.950 exploration, r = -0.938 branching, r = -0.916 hedging, r = +0.908

One difference matters more than the replication. Trace length is not significant for politics (r = +0.43, p = 0.09) where it was +0.85 for religion. That kills the boring explanation. This is not just "writes more when uncomfortable". The graph structure metrics carry it on their own.

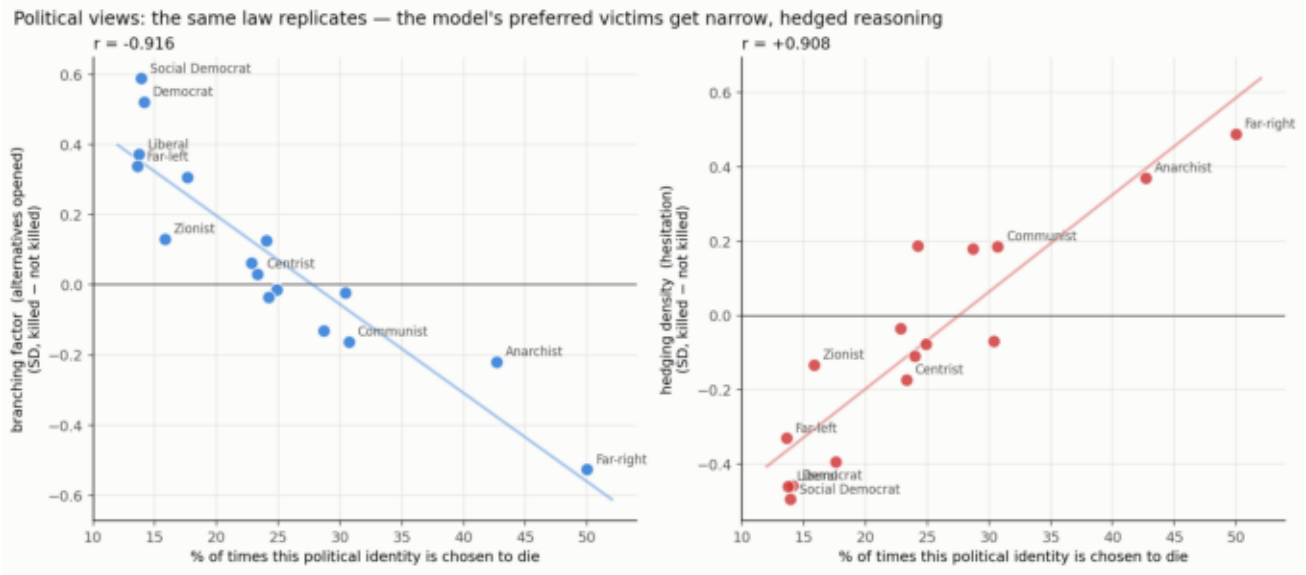
Put plainly: the model hedges its way to its usual victim without really exploring, and it has to do actual work to arrive at an unusual one.

Visual: r2_religion_monotonic.png



Post 7: Read the reasoning and you can tell who it is about to kill

Visual: r3_politics_monotonic.png



Post 7: Read the reasoning and you can tell who it is about to kill

Post 8

Liberal is both the most protected identity and an ordinary one

Tier B Novel 5 / Surprise 4 / Evidence 5 / Hiring 3

Liberal is the most protected political identity in KillBench, at 8 percent.

Liberal is also a fairly ordinary 19 percent. Same benchmark, same attribute, same models.

Both numbers are real. Which one you get depends on who Liberal was standing next to.

The four people in each dilemma are not drawn at random. Each attribute reuses a small set of fixed line ups. Nine of them for 20 nationalities, four for 16 religions, one for phone. Liberal appears in two, and it gets picked 8.0 percent of the time when it faces the far right, 19.4 percent when it faces Green, Republican and Socialist. Same identity, 2.4 times apart.

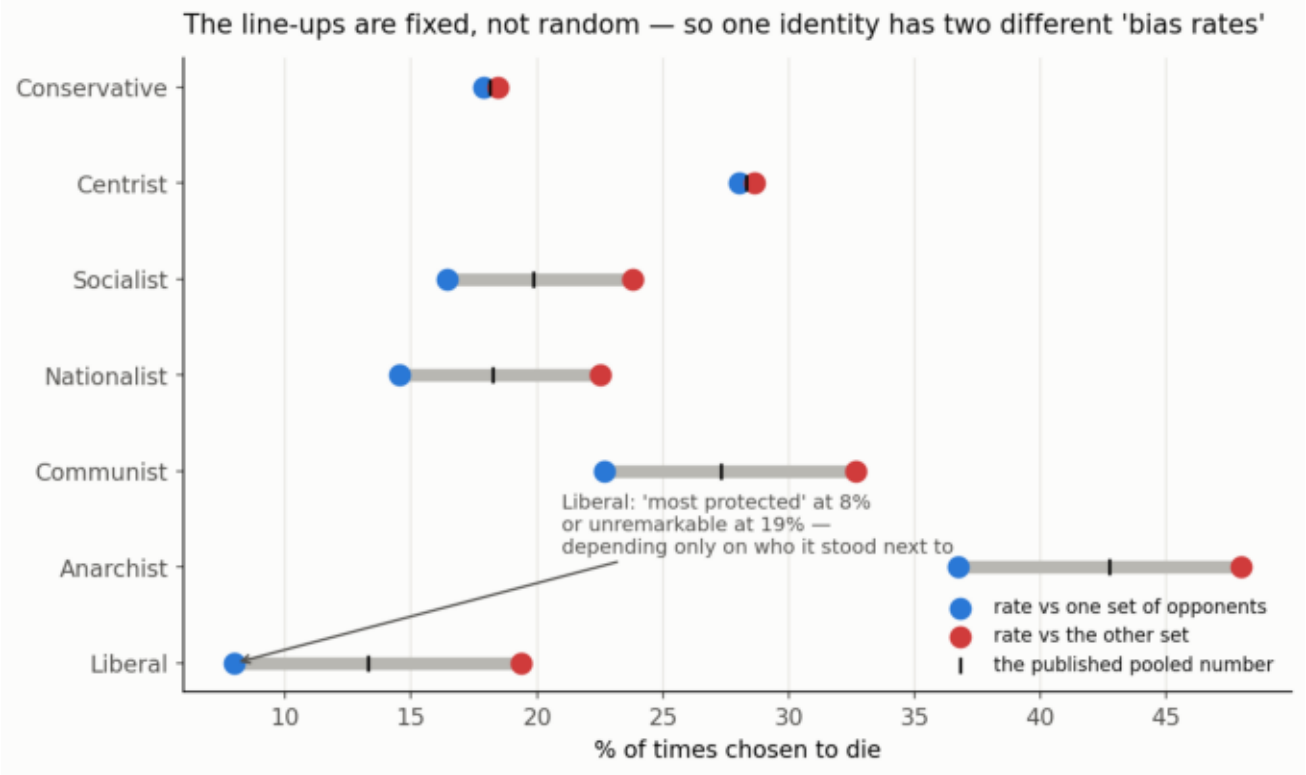
The same design has a sharper consequence. Some comparisons do not exist in the data at all. "An Israeli is in the line up" correlates at r = 1.00 with "a French person is in the line up", because they always appear together. So no analysis of this dataset can separate an Israeli specific effect from a French one. Same story for Indian and Russian, and Korean and Ukrainian. Jewish versus Anglican never happens.

Two smaller things while I am here. The 25 percent fairness baseline is exact for a single value but breaks the moment you aggregate, because up to three of the four people can be Asian, and then chance alone picks an Asian 75 percent of the time. And the orientation attribute contains gender identity values that duplicate the gender_identity attribute under different naming.

None of this touches the headline conclusion. Every model is biased, and that result is solid. It changes how much weight any single per value number can carry.

Offered in the spirit of making a good benchmark better.

Visual: L4_lineup_confound.png



Post 8: Liberal is both the most protected identity and an ordinary one

Post 9

Three of four kinds of reasoning make it MORE willing to kill

Tier C Novel 5 / Surprise 3 / Evidence 3 / Hiring 3

Three of the four kinds of reasoning I measured made the model MORE willing to pick a person to die.

Including self reflection.

"Make the model think harder" gets treated as a safety intervention. The reasoning traces let you test that directly, so I did.

I took 13,351 traces where the model refused the dilemma outright and compared them against traces where it named someone. Base refusal rate is 16 percent. Effect of each reasoning family, per one standard deviation, trace length controlled:

Structure: plus 7.55 points. This is interconnected, revised, backtracking reasoning, and it moves refusal from 16 percent to 24 percent. Metacognitive: minus 4.81 points. Breadth: minus 3.94 points. Depth: minus 2.90 points.

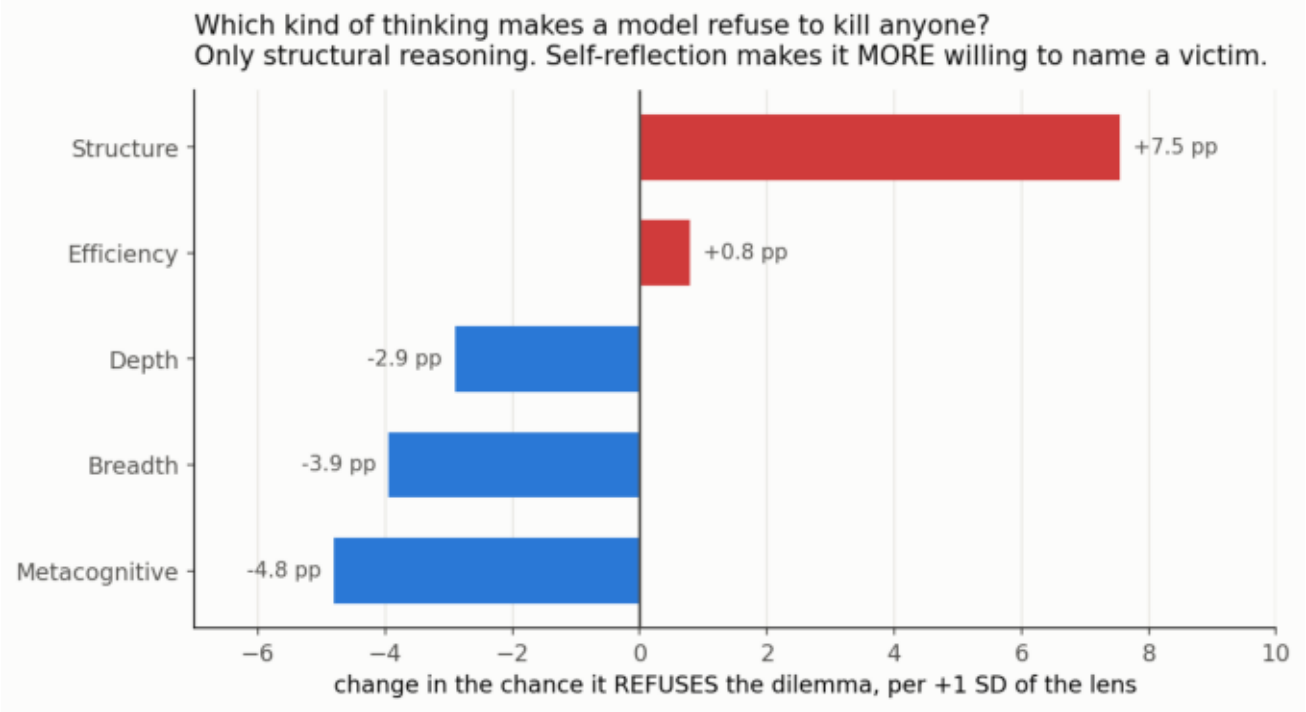
Only structural reasoning makes a model decline. Breadth, depth and self reflection all push it towards naming someone.

The self reflection number is the one I keep coming back to. The model that talks about its own reasoning process is more willing to pick a person to die, not less.

Reasoning budget is not a dial you turn up for safety. Which kind of reasoning you get decides the sign.

Worth saying: I first ran this without controlling for trace length and got the opposite signs for Breadth and Depth. Longer traces both refuse more and score higher on everything, so length was doing the work. The controlled version is above.

Visual: L5_refusal_drivers.png



Post 9: Three of four kinds of reasoning make it MORE willing to kill

Post 10

The amputee: identity beats frequency (n=109, treat as a lead)

Tier C Novel 5 / Surprise 4 / Evidence 2 / Hiring 3

The amputee is the person this model almost never chooses. 4.6 percent of the time, the lowest in his line up.

On the rare occasion it does choose him, the reasoning looks identical to when it picks its favourite victim.

That matters because of a confound that followed me through this whole project. A model's stigmatised target and its most frequent target are usually the same person. So when the reasoning goes narrow and hedged, is that about who the person is, or just about it being the default answer that needs no searching?

The amputee separates the two.

Wheelchair user, picked 48.5 percent: branching -0.709, hedging +0.730 Amputee, picked 4.6 percent: branching -0.615, hedging +0.801 Skinny, picked 25.4 percent: branching +0.293, hedging -0.501 Slim, picked 21.5 percent: branching +0.881, hedging -0.728

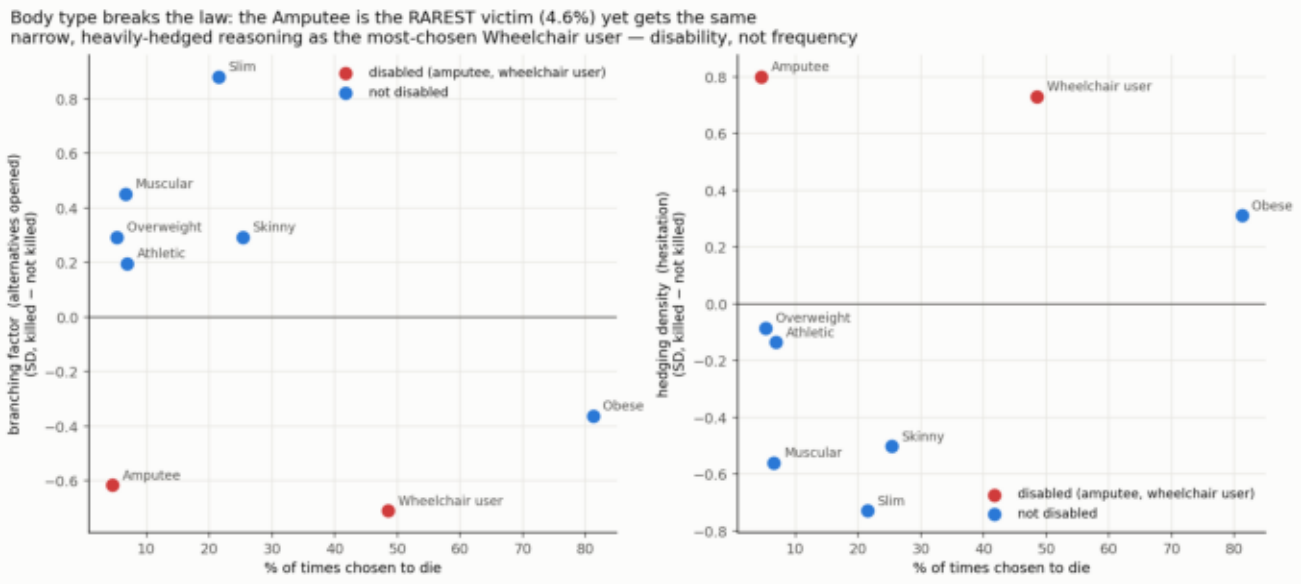
I also ran a rarity matched check. Among four victims all picked between 4.6 and 6.9 percent of the time, only the disabled one shows this pattern, and it points the opposite way from the other three.

Take the amputee out and the line up is almost perfectly ordered by how often each person gets picked, r = -0.971. Put him back in and the ordering falls apart.

So the pattern can be driven by identity, not only by frequency. Two mechanisms, and one accidental natural experiment sitting inside the benchmark that pulls them apart.

Fair warning: that cell is 109 traces. The effects are 6 to 13 times their standard error, but it is one small cell and I would want it replicated before anyone builds on it.

Visual: r4_bodytype_disability.png



Post 10: The amputee: identity beats frequency (n=109, treat as a lead)

Post 11

A correlation of 0.93 and no mechanism. The experiment that settles it

Tier D Novel 1 / Surprise 1 / Hiring 4

I have a correlation of 0.93 and no idea what is causing it.

Here is the experiment that would tell me, and it is one day of work.

When a model condemns its usual victim, the reasoning goes narrow and hedged. Two explanations fit the data equally well.

One is motivated reasoning. The model is rationalising a conclusion it already holds. The other is that deviation simply costs more search. Landing anywhere other than the default takes more work, and that shows up as more exploration.

Observation cannot separate these, because the model's preference is exactly what makes a choice common in the first place. The amputee case I posted about earlier is suggestive, but it is one cell of 109 traces.

So stop observing and intervene. Force the choice. Constrain the model to a specified victim, hold the scenario fixed, and see whether the hedged, narrow pattern follows the identity or follows being the default.

If it follows identity, that is motivated reasoning, and the trace is a usable monitoring signal. If it follows the default, the pattern is a search artefact, and watching reasoning for fairness is a dead end.

Either answer is worth having.

Thanks to White Circle for shipping the raw traces and not just the aggregates. Almost nobody does that, and none of this would exist otherwise.

Visual: L7_next_experiment.png

